

Lecture 5-1: atomic shells and classical models of chemical bonding-3

Jin Xie (谢进), SPST, ShanghaiTech

Email: xiejin@shanghaitech.edu.cn

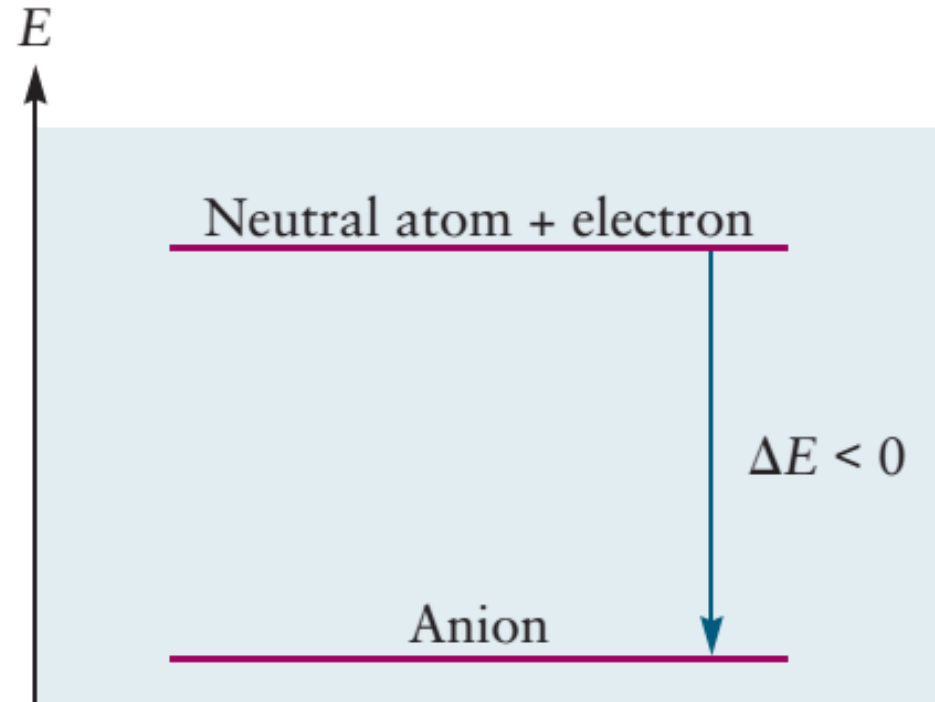
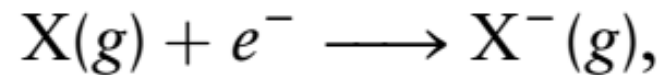
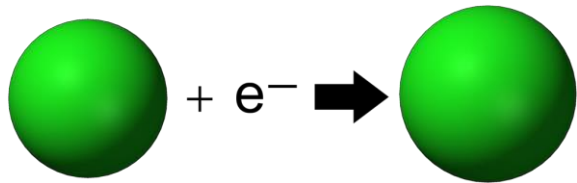
Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

2025/10/15

Electron affinity

Electron attachment energy (电子附着能): an atom accepts an extra electron to form an anion.

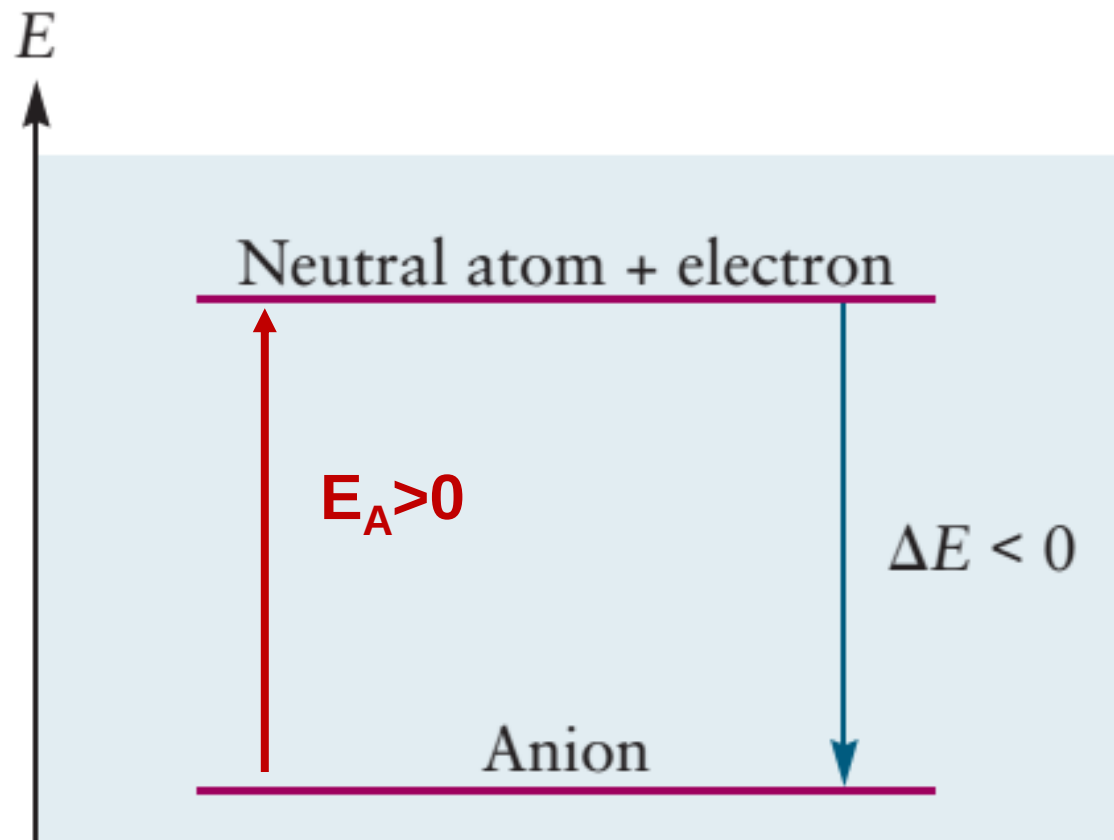
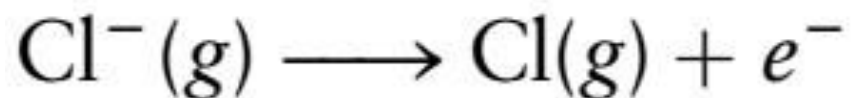


Usually, $\Delta E < 0$

Difficult to measure electron attachment energy!

Electron affinity

Electron affinity (EA_x) (电子亲和能) of atom X: energy required to detached the electron from the anion X^- and give the neutral atom.

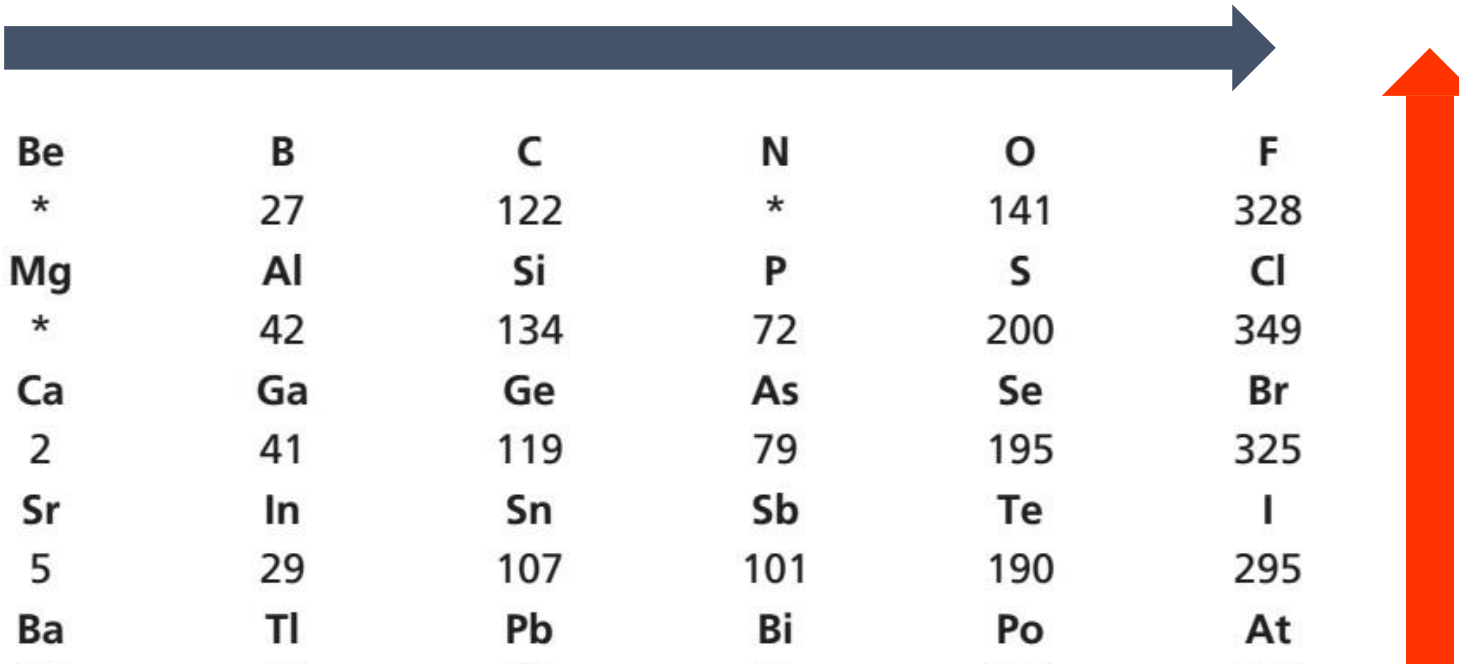


$$EA_{Cl} = 349 \text{ kJ/mol}$$

Electron affinity

TABLE 3.2

Electron Affinity of Selected Atoms (in kJ mol^{-1})



H								He
73								*
Li	Be	B	C	N	O	F		Ne
60	*	27	122	*	141	328		*
Na	Mg	Al	Si	P	S	Cl		Ar
53	*	42	134	72	200	349		*
K	Ca	Ga	Ge	As	Se	Br		Kr
48	2	41	119	79	195	325		*
Rb	Sr	In	Sn	Sb	Te	I		Xe
47	5	29	107	101	190	295		*
Cs	Ba	Tl	Pb	Bi	Po	At		Rn
46	14	19	35	91	183	270		*

*No stable anion A^- exists for this element in the gas phase.

No gaseous atom has a positive electron affinity for a second electron.

Electronegativity

Electronegativity (电负性): the tendency of atoms to attract electrons in molecules.

Mulliken's Electronegativity

$$\text{EN (Mulliken)} \propto \frac{1}{2} (IE_1 + EA)$$



Robert C. Mulliken
(1896–1986)

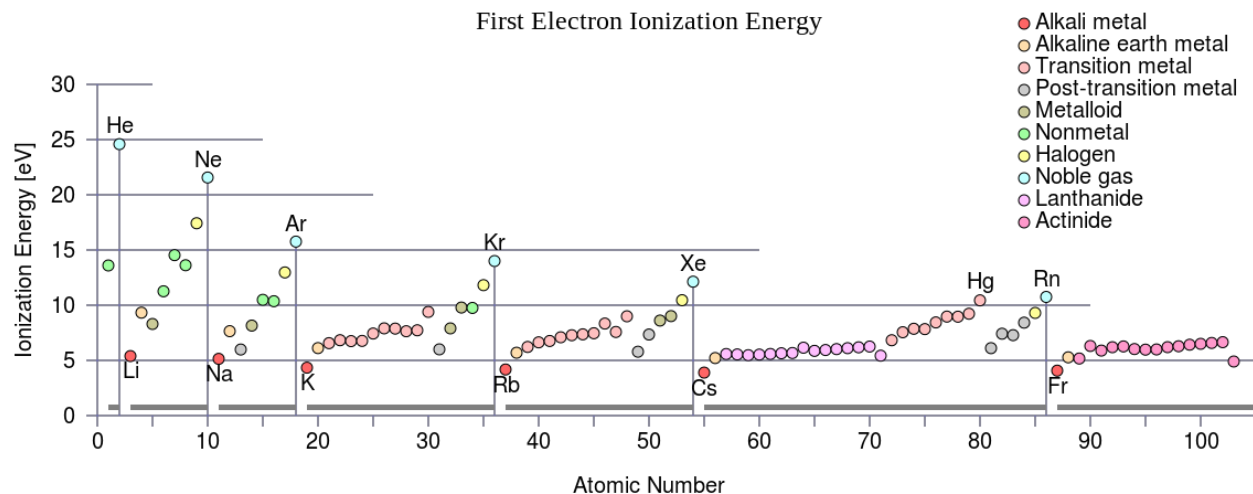


TABLE 3.2

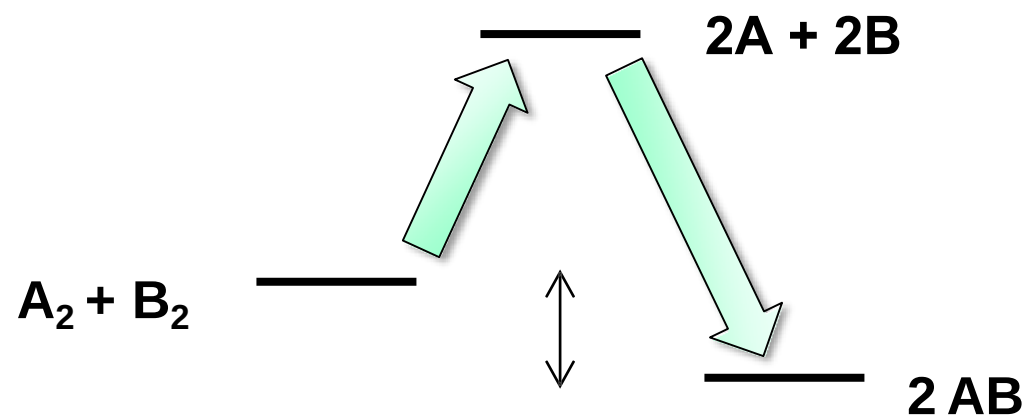
Electron Affinity of Selected Atoms (in kJ mol⁻¹)

H							He
73							*
Li	Be	B	C	N	O	F	Ne
60	*	27	122	*	141	328	*
Na	Mg	Al	Si	P	S	Cl	Ar
53	*	42	134	72	200	349	*
K	Ca	Ga	Ge	As	Se	Br	Kr
48	2	41	119	79	195	325	*
Rb	Sr	In	Sn	Sb	Te	I	Xe
47	5	29	107	101	190	295	*
Cs	Ba	Tl	Pb	Bi	Po	At	Rn
46	14	19	35	91	183	270	*

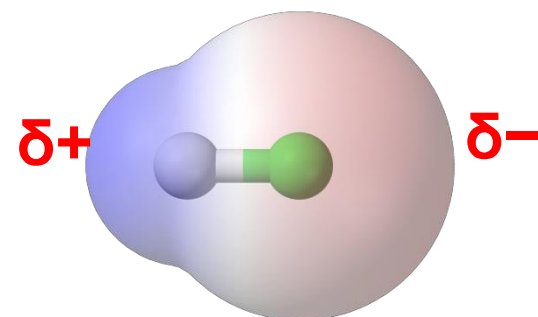
*No stable anion A⁻ exists for this element in the gas phase.

Pauling's Electronegativity

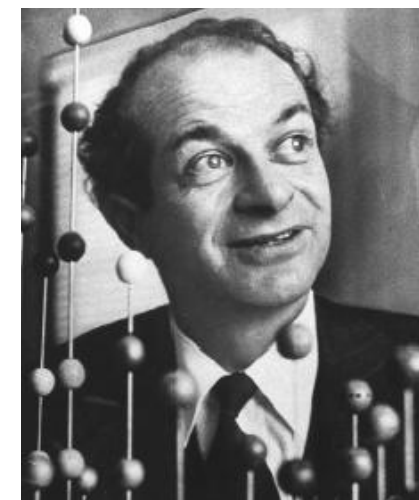
- NaCl vs. HF, HCl, HBr vs. H₂, F₂, Cl₂, Br₂



$$2 \times E_{\text{bond}} (\text{H-Cl}) > E_{\text{bond}} (\text{H-H}) + E_{\text{bond}} (\text{Cl-Cl})$$



$$\Delta \propto q^2$$



Linus Pauling
(1901–1994)

The extra stability was provided by partial **charge separation** in the bond, called **ionic character**.

Pauling's Electronegativity

Pauling's Electronegativity scale: a **comparison of the bond energies** of a large number of **heteronuclear bond pairs** with those of **homonuclear diatomics** comprising the same elements.

An empirical formula:

1. Dissociation energy of A-A bond and B-B bond (both are covalent bonds): ΔE_{AA} , ΔE_{BB}
2. Dissociation energy of A-B bond (only **covalent contribution part**): $(\Delta E_{AA} \times \Delta E_{BB})^{0.5}$
3. Dissociation energy of A-B bond (**covalent contribution + ionic contribution**): ΔE_{AB}

Excess bond energy $\Delta = \Delta E_{AB} - \sqrt{\Delta E_{AA} \Delta E_{BB}}$

Pauling's definition of **electronegativity difference**

$$\chi_A - \chi_B = 0.102 \Delta^{1/2}$$

Pauling's Electronegativity

Pauling arbitrarily assigned the value 4.0 to F.

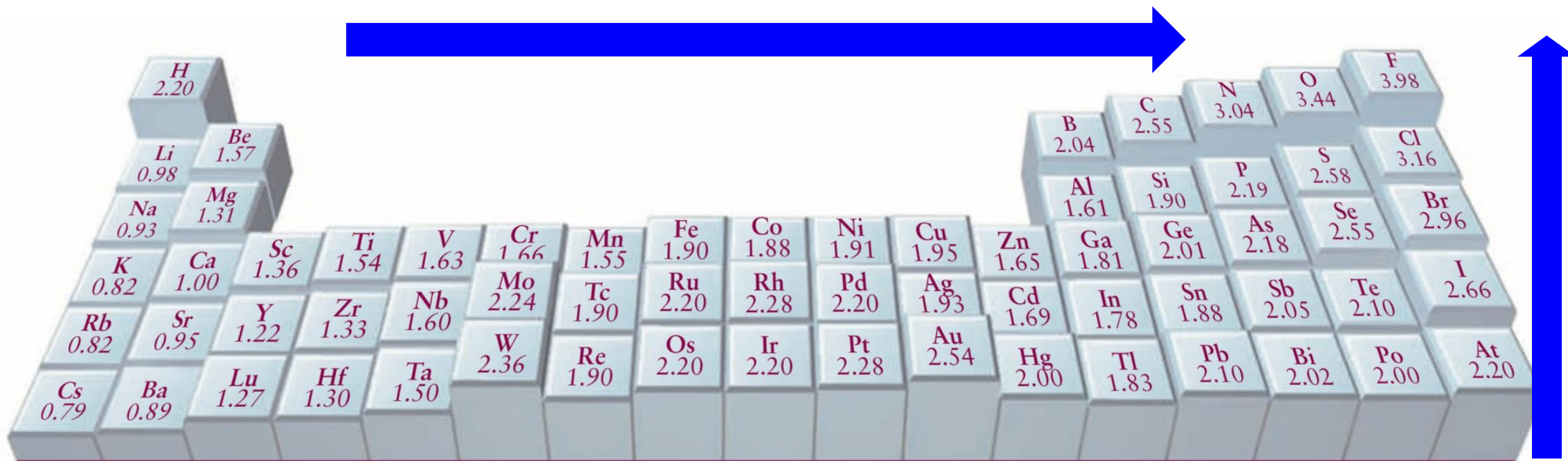
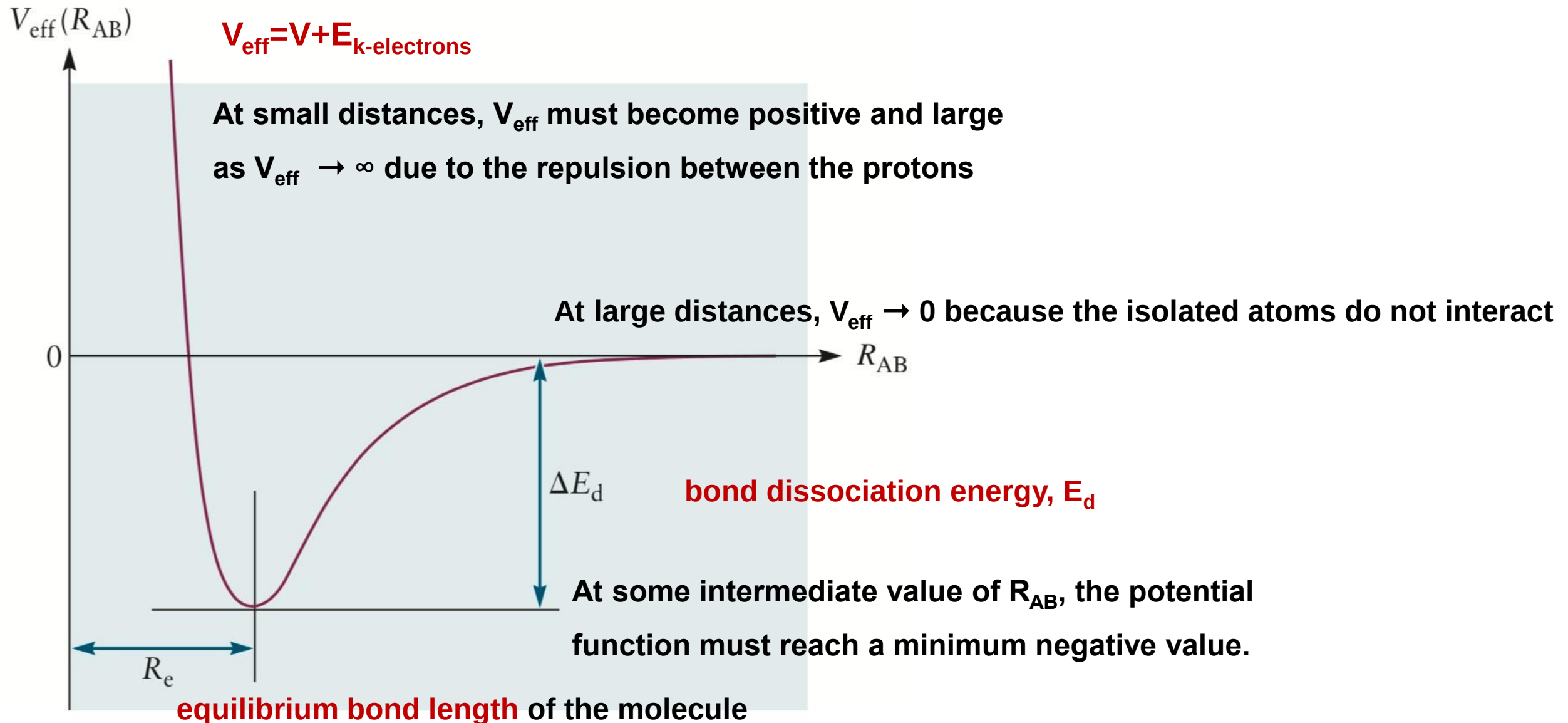


FIGURE 3.7 Average electronegativity of atoms, computed with the method that Linus Pauling developed. Electronegativity values have no units.

Forces and potential energy in molecules

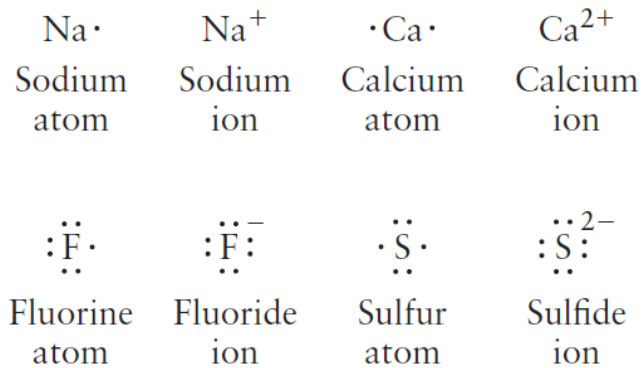


Ionic bonding (离子键)

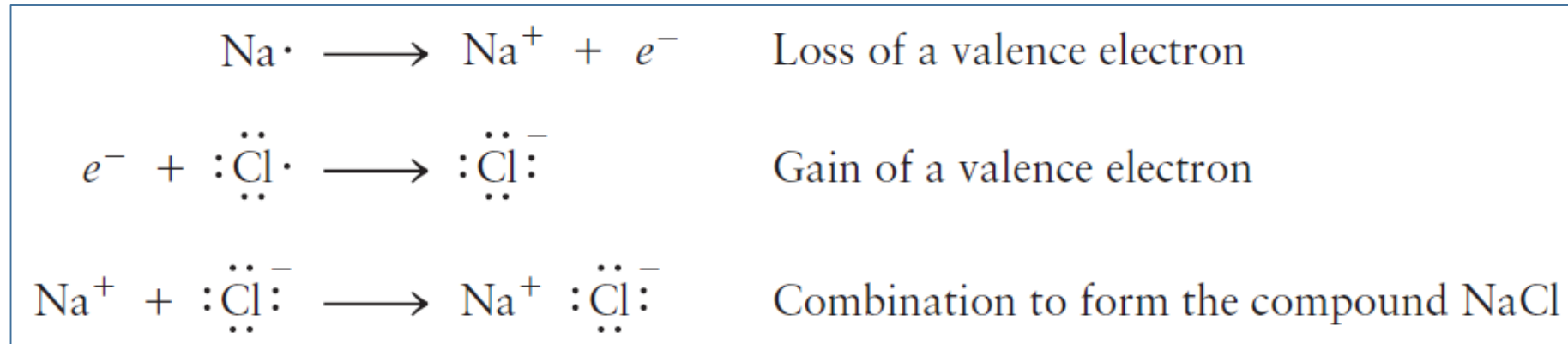
An **ionic bond** forms between atoms with large differences in electronegativity, such as Na and F.



The more electropositive atom **transfers an electron** to the more electronegative atom, forming an ion pair that is then drawn together by the attractive electrostatic force.

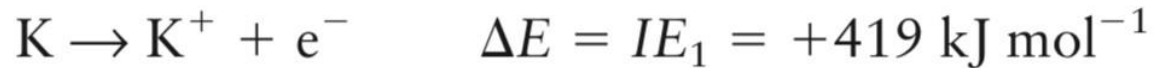


Lewis symbols

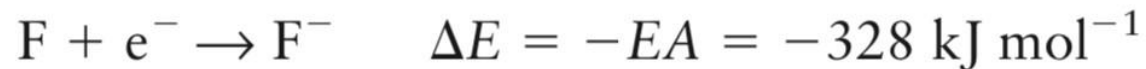


ionic compound

Ionic bonding



$$\Delta E_{\infty} = IE_1(\text{K}) - EA(\text{F}) = +91 \text{ kJ mol}^{-1}$$



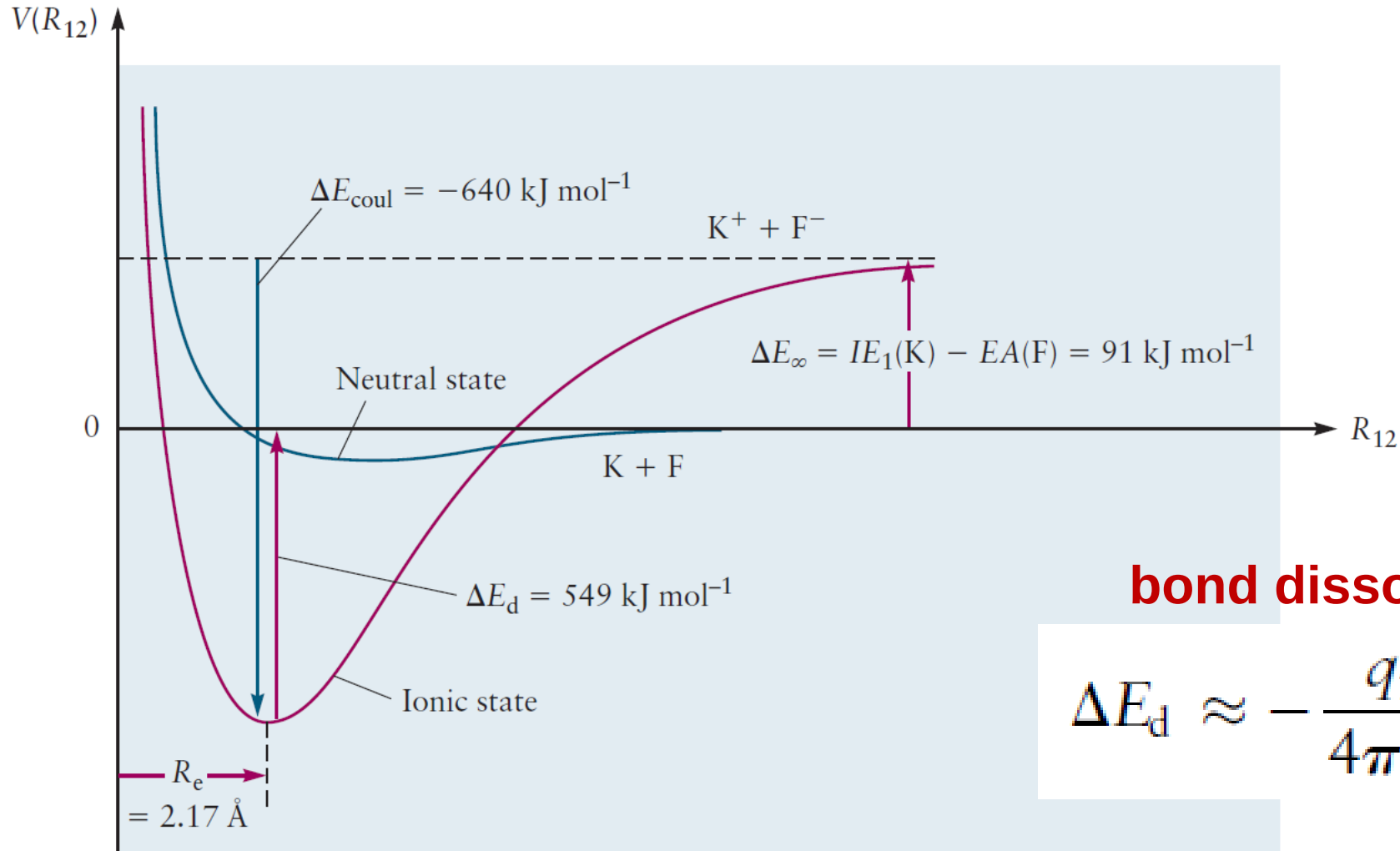
The total energy cost for the creation of this ion pair is always **positive**:
the smallest ionization energy of any element > the largest electron affinity of any element.

$$V(R_{12}) = \frac{q_1 q_2}{4\pi\epsilon_0 R_{12}} \text{ (J per ion pair)}$$

$$V(R_{12}) = \frac{q_1 q_2}{4\pi\epsilon_0 R_{12}} \cdot \frac{N_A}{10^3} \text{ (kJ mol}^{-1}\text{)}$$

The ions are attracted to one another by the electrostatic force, which is described by Coulomb's law. This is called **Coulomb stabilization energy**.

Ionic bonding



bond dissociation energy

$$\Delta E_d \approx -\frac{q_1 q_2}{4\pi\epsilon_0 R_e} \frac{N_A}{10^3} - \Delta E_\infty$$

FIGURE 3.13 The potential energy of the ions K^+ and F^- , and the atoms K and F , as a function of their internuclear separation R_{12} .

Covalent bonding (共价键)

What kinds of bonds are formed between elements of identical or comparable electronegativities (e.g. H₂ or CO)?



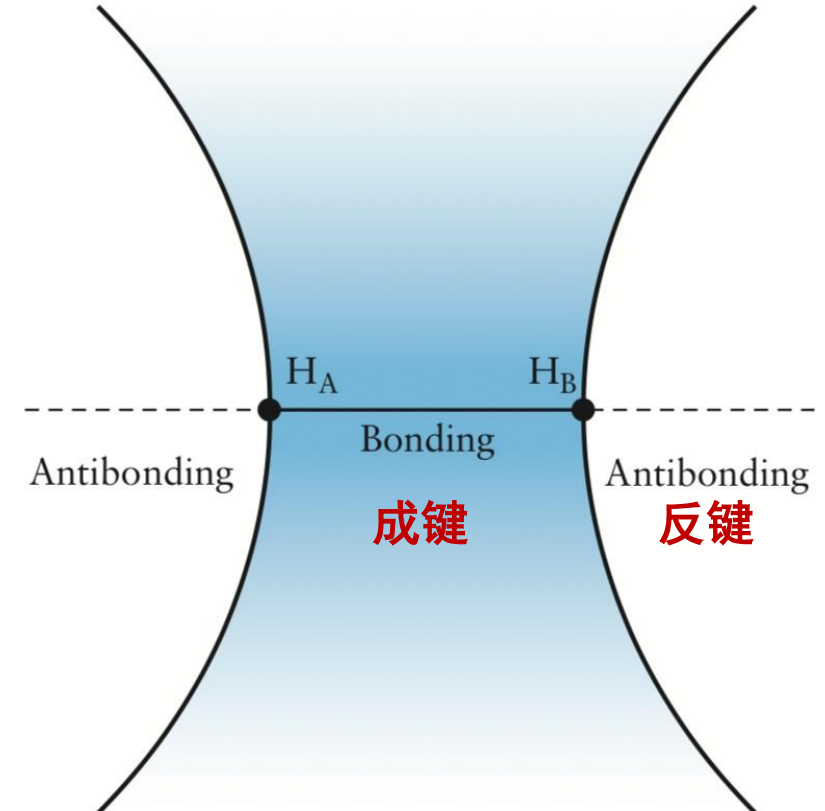
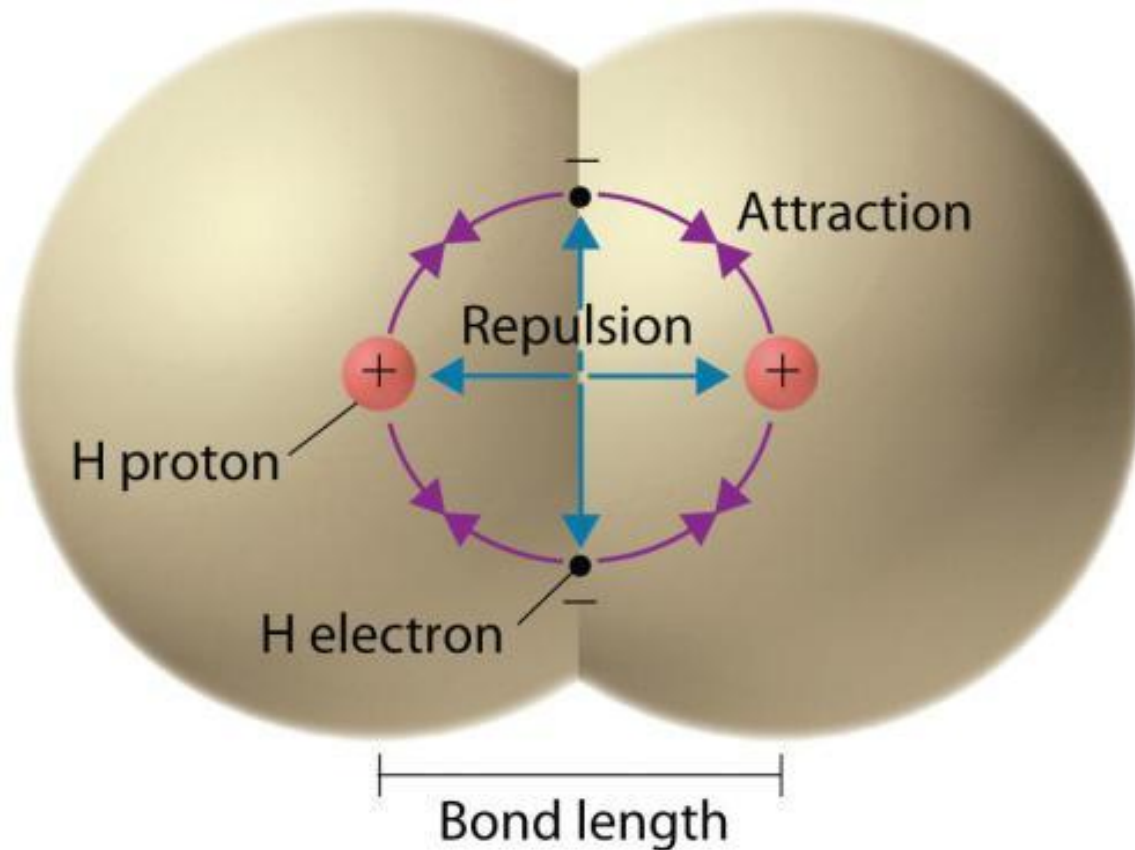
No Coulomb stabilization energy from ions with opposite charges



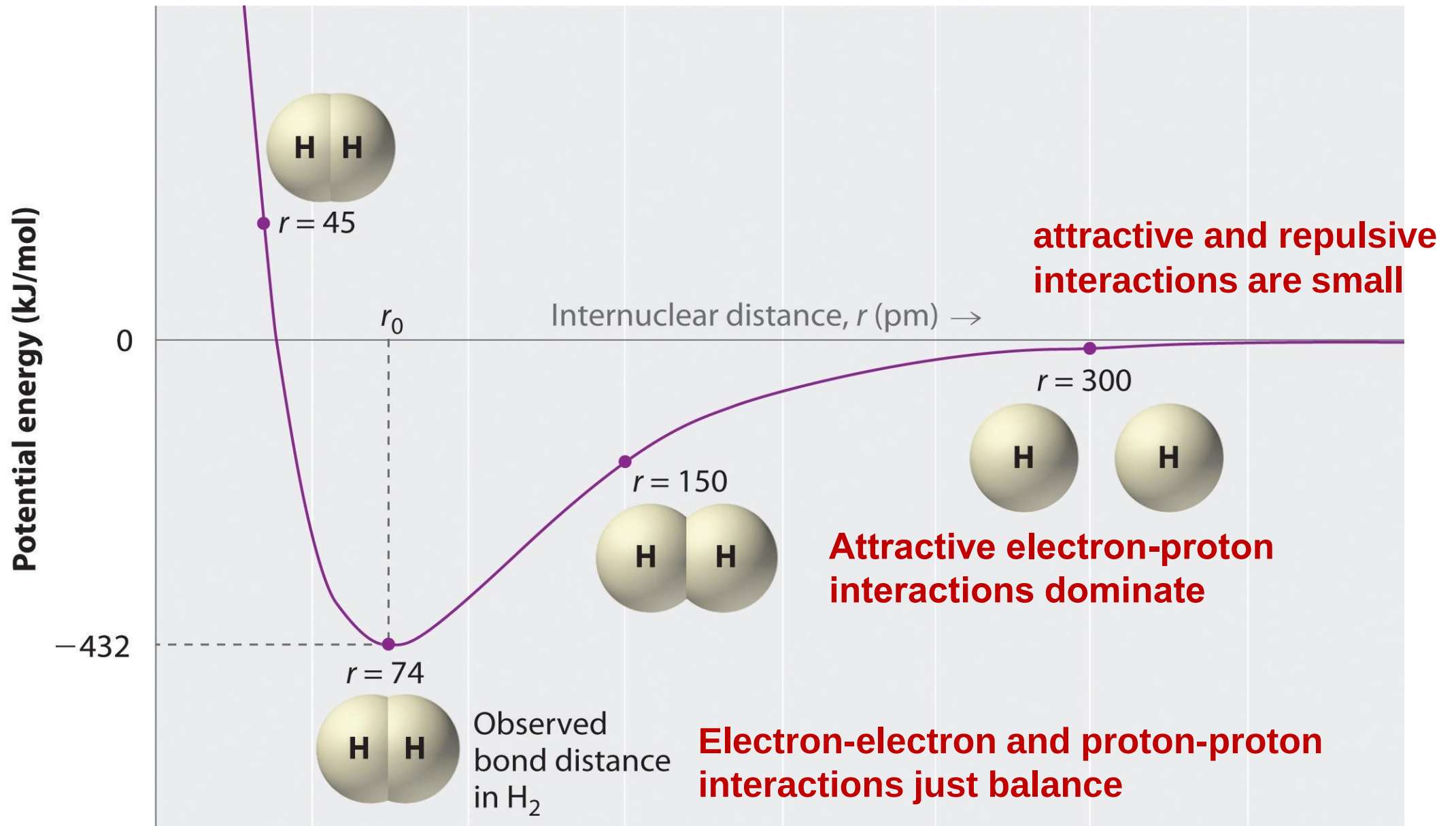
What is the **driving force** for bond formation from separated atoms in the gas phase?

Covalent bonding

- This simple model is supported by experimental data for H_2^+ .
- $R_e = 1.06 \text{ \AA}$, $E_d = 255.5 \text{ KJ mol}^{-1}$, which is characteristic of a stable covalent bond.



Covalent bonding

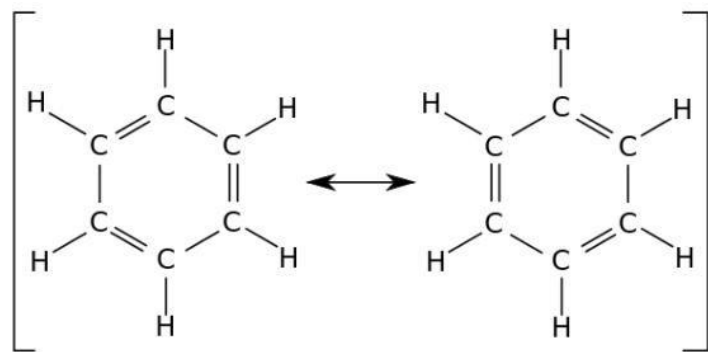


Covalent bonding

- **Bond order (键级):** the number of shared electron pairs between the two atoms.
 - 键级是衡量化学键相对强弱的参数，键级愈大，键愈稳定
 - 若键级为零，则不能成键。

TABLE 3.5 Three Types of Carbon–Carbon Bonds

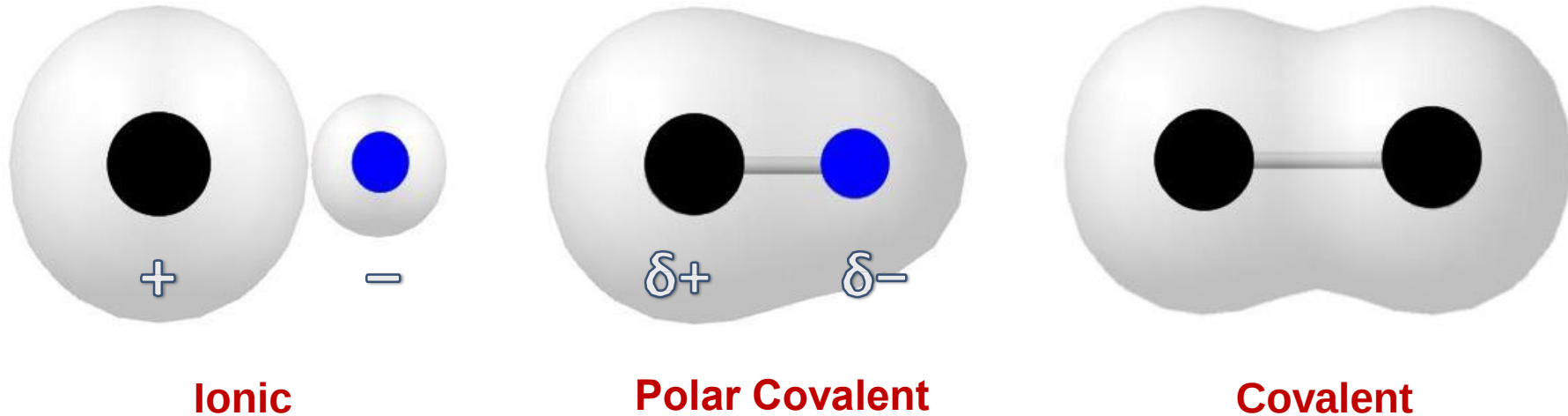
Bond	Molecule	Bond Length (Å)	Bond Energy (kJ mol ⁻¹)
C—C	C ₂ H ₆ (or H ₃ CCH ₃)	1.536	345
C=C	C ₂ H ₄ (or H ₂ CCH ₂)	1.337	612
C≡C	C ₂ H ₂ (or HCCH)	1.204	809



1.397 Å

505 kJ mol⁻¹

Polar covalent bonding (极性共价键)



- **Polar Covalent:** bonds in which there is a partial transfer of charge
- The absolute value of the **difference in electronegativity** of two bonded atoms tells the **degree of polarity** in their bond.

Ionic	Polar Covalent	Covalent
$\Delta\chi > 2.0$		$\Delta\chi < 0.4$

Dipole moments and percent ionic character

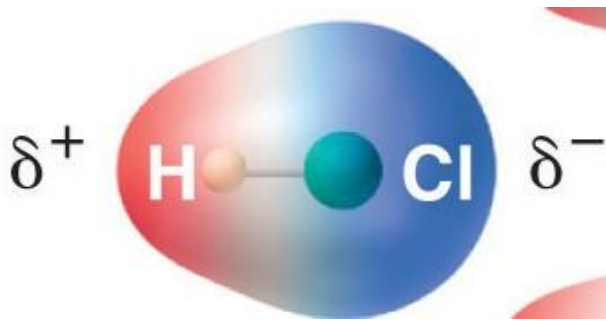
If two charges of equal magnitude and opposite sign, $+q$ and $-q$, are separated by a distance R , the **dipole moment μ (偶极矩)** of that charge distribution is:

$$\mu = qR$$

debye (D), $1\text{D} = 3.336 \times 10^{-30} \text{ C m}$



The debye can also be defined as the dipole moment of two charges $\pm e$ separated by $0.2082 \text{ \AA}$.



$$\mu(\text{D}) = [R(\text{\AA})/0.2082 \text{ \AA}] \delta$$

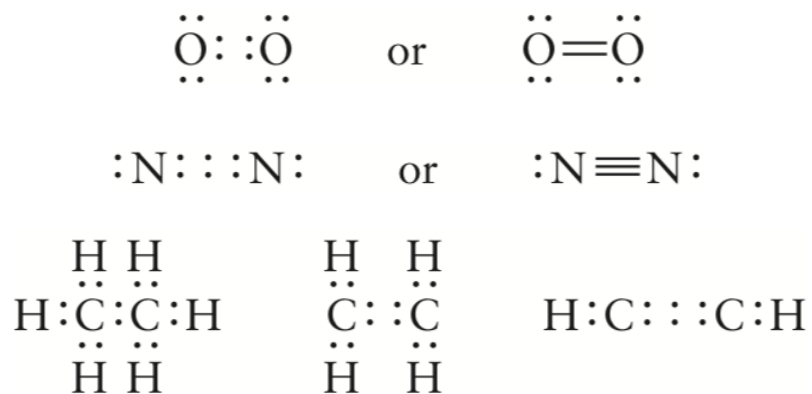
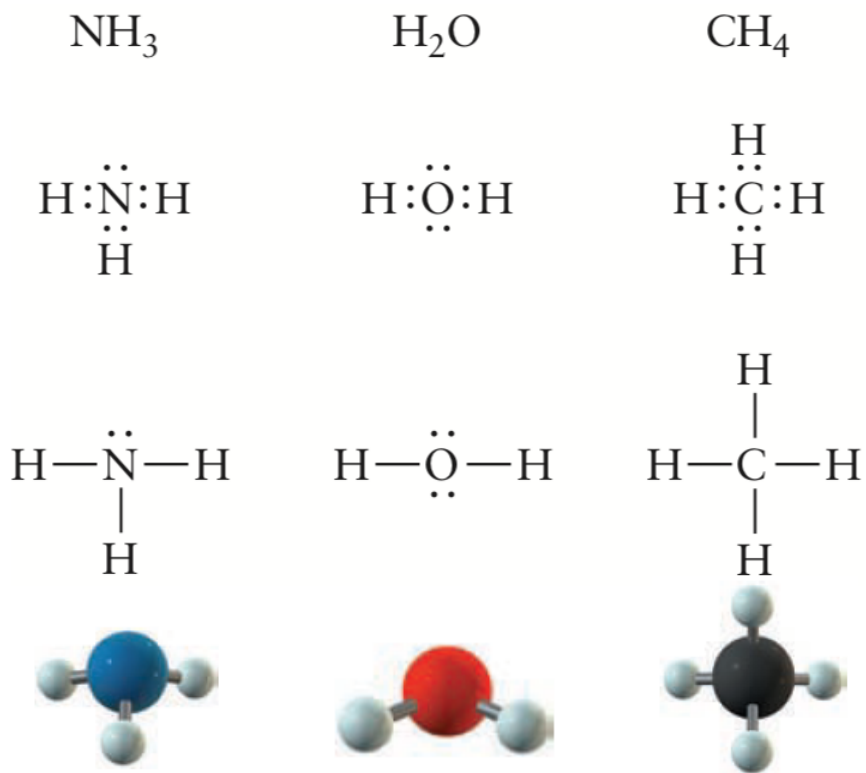
If δ is the fraction of a unit charge on each atom in a diatomic molecule ($q = e\delta$) and R is the equilibrium bond length

δ is the fraction ionic character

Lewis diagrams

Octet rule (八隅律) : Whenever possible, the electrons in a covalent compound are distributed in such a way that each main-group element (except H) is surrounded by eight electrons.

- A **shared pair electrons** = a **covalent bond**
- A **lone pair electrons** = **no bonding**
- Each atom achieves its own **noble-gas shell** of electrons
- more than one pair may be shared (**double bond, triple bond**)



Lewis diagrams: formal charges

Formal charge (形式电荷): the charge an atom in a molecule would have if the electrons in its Lewis diagram were divided equally among the atoms that share them.



Formal charge = number of valence electrons

- number of electrons in lone pairs**
- $\frac{1}{2}$ (number of electrons in bonding pairs)**

How to draw Lewis diagrams

Step 1: calculate total number of valence electrons + negative ion - positive ion



$$5 + 6 + 3 \times 7 = 32$$

Step 2: calculate number of electrons needed to form noble-gas shell

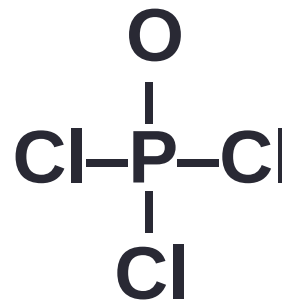
$$5 \times 8 = 40$$

Step 3: calculate numbers of shared electrons = step 2 - step 1

$$40 - 32 = 8$$

Step 4: assign two bonding electrons to each bond

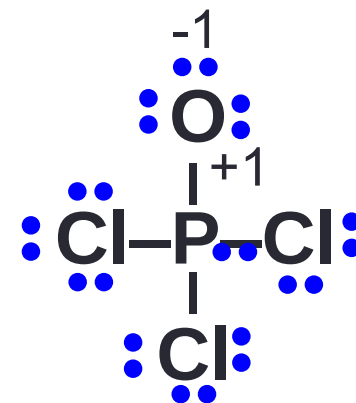
Step 5: assign remaining bonding electrons to make double or triple bonds



Step 6: assign remaining electrons as lone pairs to the atoms, giving octets to all atoms except H.

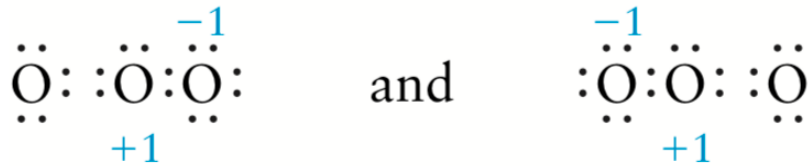
Step 7: Determine the formal charge on each atom and write it next to that atom.

Step 8: If more than one diagram is possible, choose the one with the smallest magnitudes of formal charges and with any negative formal charges placed on the most electronegative atoms



Lewis diagrams: resonance forms

For certain molecules or molecular ions, two or more **equivalent Lewis diagrams** can be written



Are there any difference between two O-O and O=O bonds?

The two O-O bond lengths are found experimentally to be identical (1.28 Å), which is intermediate between the O-O single bond length in H₂O₂ (1.49 Å) and the O=O double bond length in O₂ (1.21 Å).



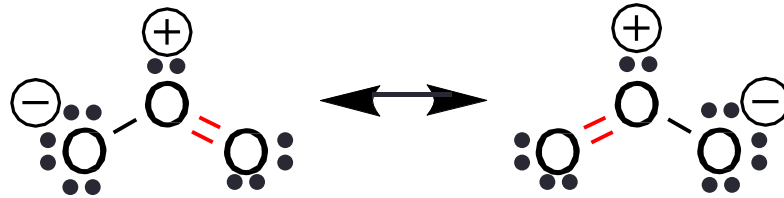
resonance hybrid



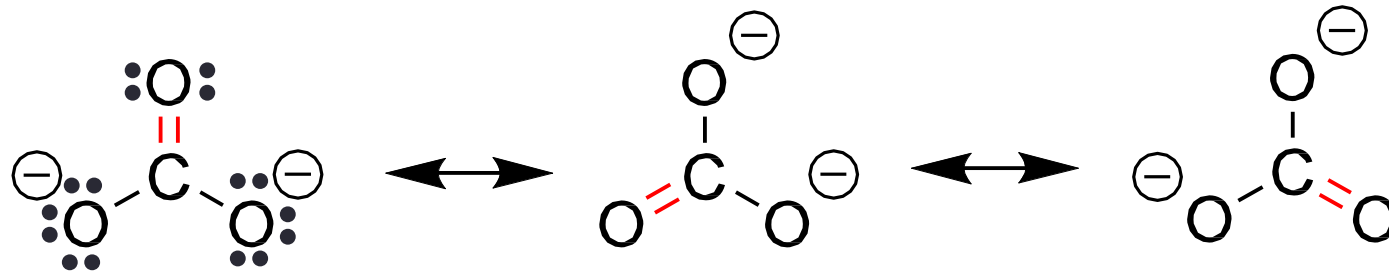
The term **resonance** (共振) does not mean that the molecule physically oscillates. The best representation of the actual bonding is a hybrid diagram that includes features of each of the acceptable individual diagrams.

Lewis diagrams: resonance forms and Bond Order

$$\text{B.O. (Resonance hybrid)} \approx \overline{\text{B.O. (Canonical forms)}}$$



O–O bond order = 1.5



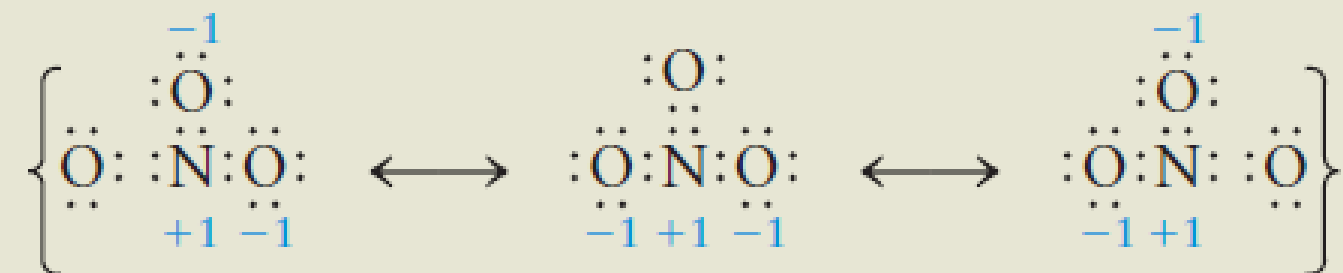
C–O bond order $\approx$ 1.3



Draw three resonance forms for the nitrate ion NO_3^- , and estimate the bond lengths.

Solution

NO_3^- contains 24 valence electrons. For each atom to have its own octet, $4 \times 8 = 32$ electrons would be required. Therefore, $32 - 24 = 8$ electrons must be shared between atoms, implying a total of four bonding pairs. These can be distributed in one double and two single bonds, leading to the equivalent resonance diagrams:



The two singly bonded oxygen atoms carry formal charges of -1 , and the charge of the nitrogen atom is $+1$. The bond lengths should all be equal and should lie between the values given in Table 3.5 for $\text{N}-\text{O}$ ($1.43 \text{ \AA}$) and $\text{N}=\text{O}$ ($1.18 \text{ \AA}$). The experimentally measured value is $1.24 \text{ \AA}$.

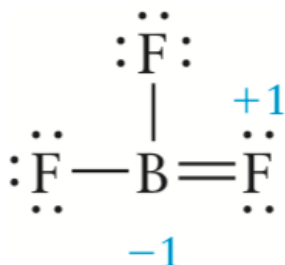
Lewis diagrams: breakdown of the octet rule

Case 1: odd-electron molecules

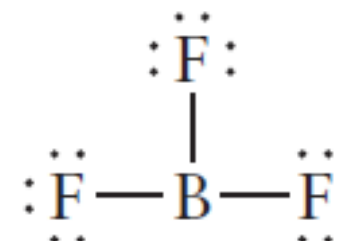


Nitrogen oxide has 11 electrons, and this is the best electron dot diagram.

Case 2: octet-deficient molecules



Experimental evidence suggests
there are no double bonds in BF_3

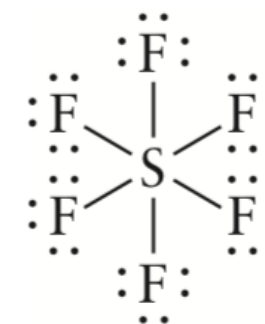


Case 3: valence shell expansion

$$56(7 \times 8) - 48(6 + 6 \times 7) = 8 \text{ bonding electrons}$$



Step 6': abandon the octet rule for the central atom but preserve it for the terminal atoms. If any electrons still remain, assign them to the central atoms as lone pairs if the central atoms is not one from the 2nd period

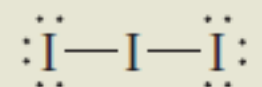


Write a Lewis diagram for I_3^- , the linear (tri-iodide) ion.

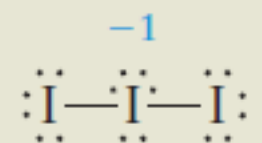
Solution

There are 7 valence electrons from each iodine atom plus 1 from the overall ion charge, giving a total of 22. Because $3 \times 8 = 24$ electrons would be needed in separate octets, $24 - 22 = 2$ are shared according to the original rules. Two electrons are not sufficient to make two different bonds, however, so valence expansion is necessary.

A pair of electrons is placed in each of the two bonds, and rule 6' is used to complete the octets of the two terminal I atoms. This leaves



At this stage, 16 valence electrons have been used. The remaining 6 are placed as lone pairs on the central I atom. A formal charge of -1 then resides on this atom:

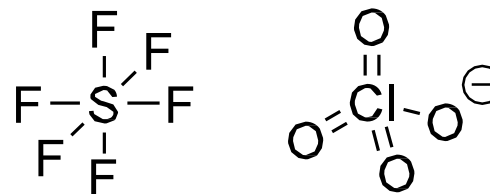


Note the valence expansion on the central atom: It shares or owns a total of 10 electrons, rather than the 8 required by adherence to the octet rule.

Summary

Molecular Orbital	Lewis Dot Structure
Bonding orbital	Shared electron pair
Nonbonding orbital	Lone electron pair
Delocalized orbital	Resonance structures
Two electrons from one atom	Coordinate bond
	Formal charge

- The **octet rule** works best for **C, N, O, F**.

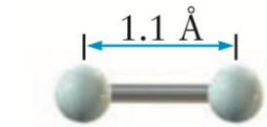


Major points of today's lecture

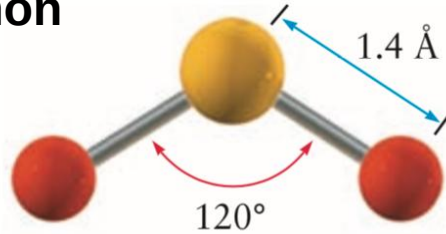
- **Atomic shells and classical models of chemical bonding**
 1. Forces and potential energy in molecules: formation of chemical bonds
 2. Ionic bonding
 3. Covalent and polar covalent bonding
 4. Electron pair bonds and Lewis diagrams for molecules
 5. The shapes of molecules
 6. Oxidation numbers
 7. Inorganic nomenclature

The shapes of molecules

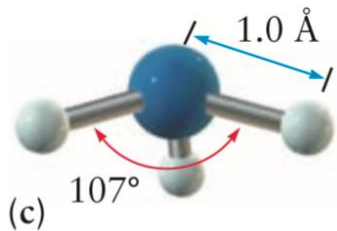
- **Bond length:** distance between the atomic nuclei in a particular bond
- **Bond angle:** angle between the axes of adjacent bonds
- **Dihedral angle:** angle between two planes, each defined by three atoms with the planes having two atoms in common



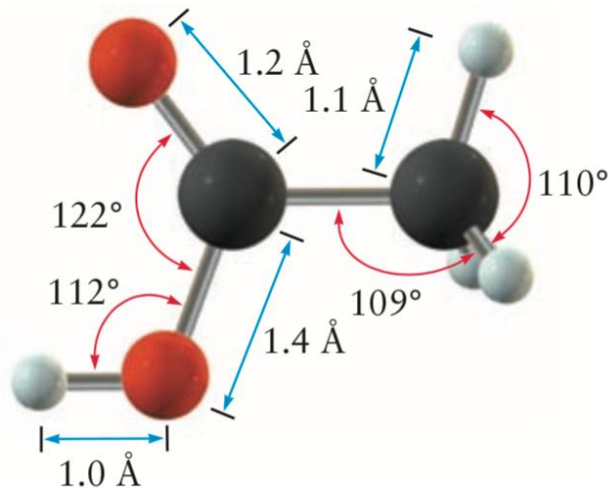
(a)



(b)



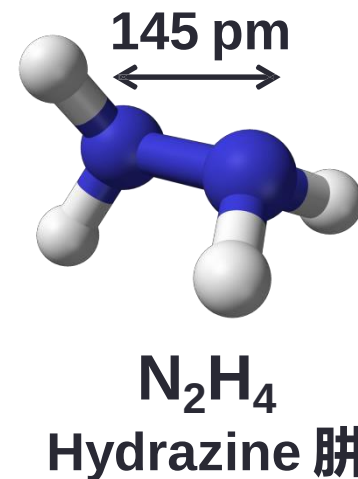
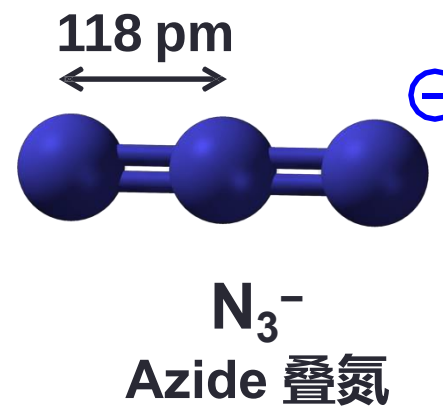
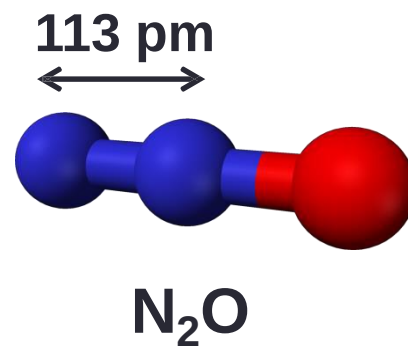
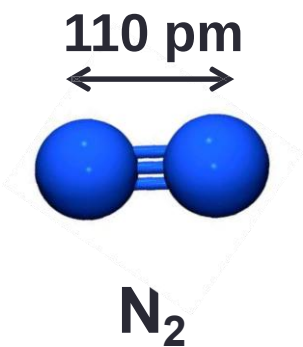
(c)



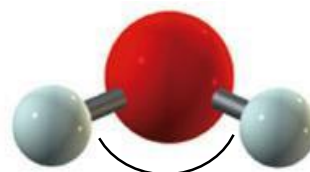
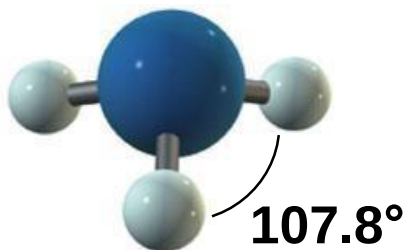
(d)

Molecular shape or geometry is governed by energetics; a molecule assumes the geometry that gives it the **lowest potential energy**.

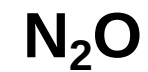
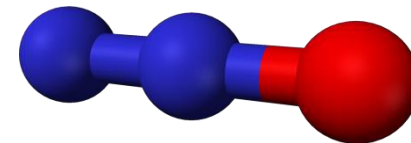
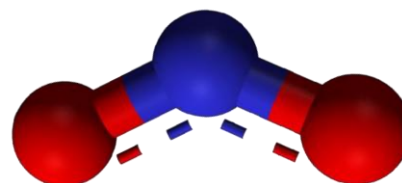
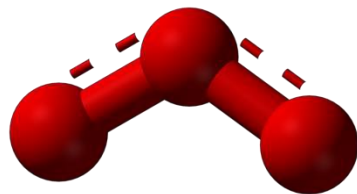
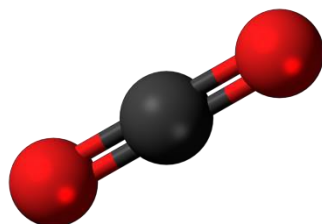
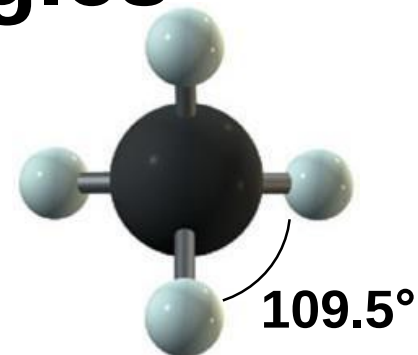
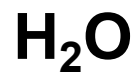
Polyatomic molecules: bond lengths



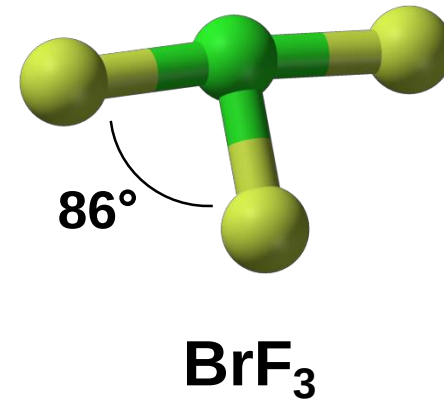
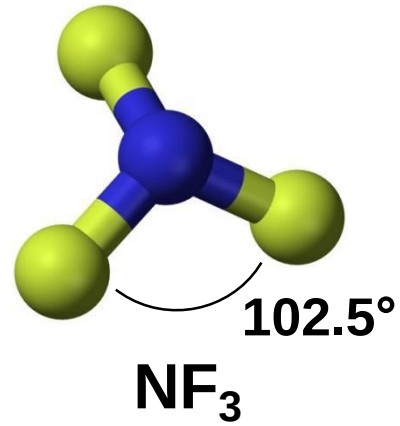
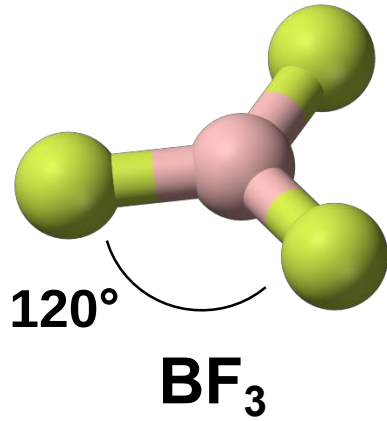
Polyatomic molecules: bond angles



104.5°



Polyatomic molecules: bond angles



The Valence Shell Electron-Pair Repulsion Theory (VSEPR)

- The Valence Shell Electron-Pair Repulsion Theory (VSEPR): 价层电子对互斥模型



- fundamental idea: the electron pairs position themselves as far apart as possible to minimize their repulsions



$$SN = \left(\begin{array}{c} \text{number of atoms} \\ \text{bonded to central atom} \end{array} \right) + \left(\begin{array}{c} \text{number of lone pairs} \\ \text{on central atom} \end{array} \right)$$

SN: steric number (空间位数)



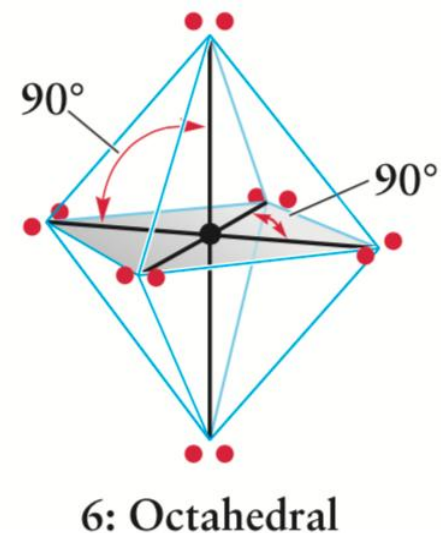
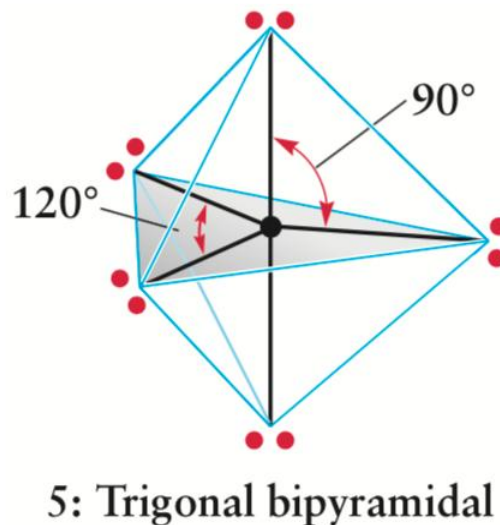
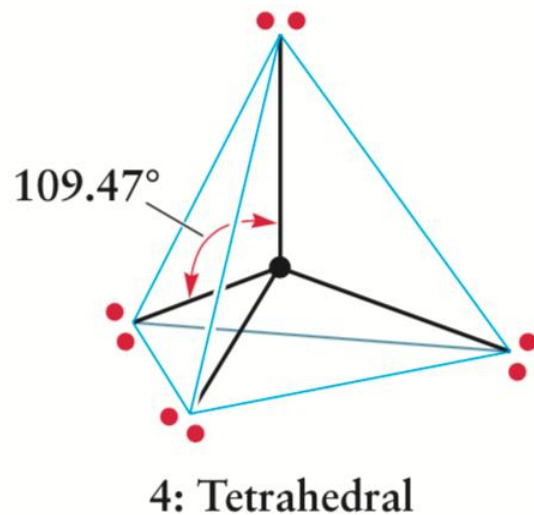
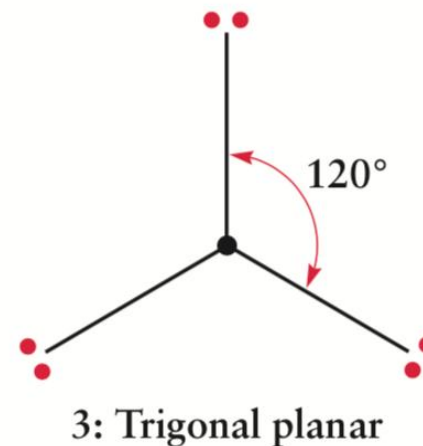
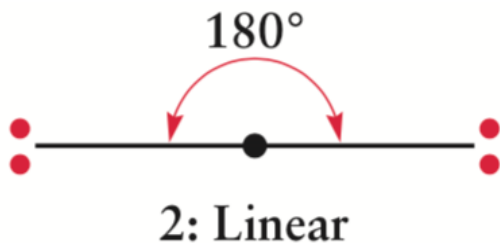
Ronald J. Gillespie
(1924–)



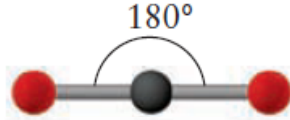
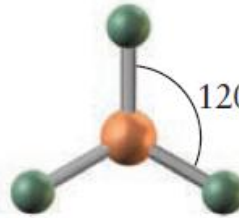
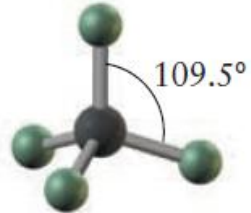
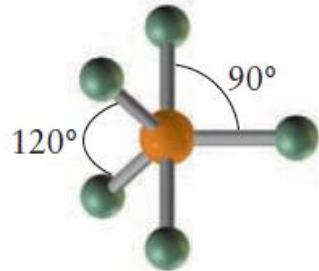
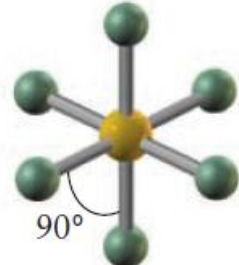
Ronald S. Nyholm
(1917–1971)

The Valence Shell Electron-Pair Repulsion Theory (VSEPR)

SN: steric number (空间位数)



Minimum energy for symmetrical molecules

AX_2	2	Linear	 A diagram of a linear molecule with a central black atom bonded to two red atoms. A curved line indicates a bond angle of 180°.	CO_2
AX_3	3	Trigonal planar	 A diagram of a trigonal planar molecule with a central orange atom bonded to three green atoms. A curved line indicates a bond angle of 120°.	BF_3
AX_4	4	Tetrahedral	 A diagram of a tetrahedral molecule with a central black atom bonded to four green atoms. A curved line indicates a bond angle of 109.5°.	CF_4
AX_5	5	Trigonal bipyramidal	 A diagram of a trigonal bipyramidal molecule with a central orange atom bonded to five green atoms. Two curved lines indicate bond angles of 90° and 120°.	PF_5
AX_6	6	Octahedral	 A diagram of an octahedral molecule with a central yellow atom bonded to six green atoms. Two curved lines indicate bond angles of 90°.	SF_6

The Valence Shell Electron-Pair Repulsion Theory (VSEPR)

Step 1: Rewrite a structure as AX_mE_n , m = number of ligands, n = number of lone pairs

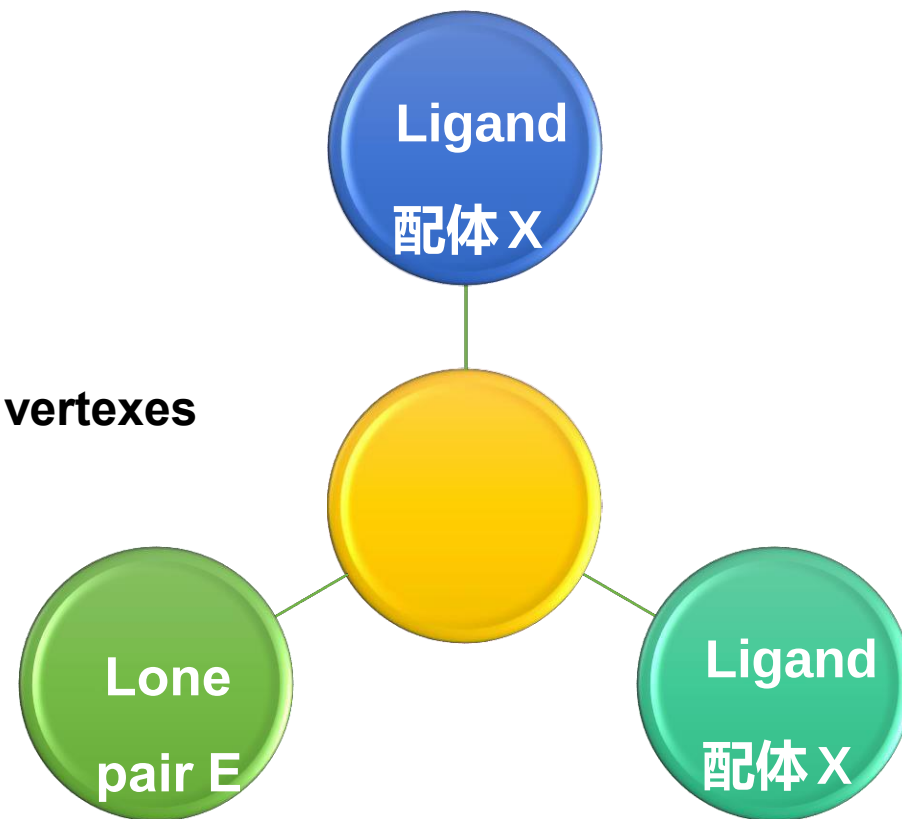
Step 2: Assign a Steric Number; $SN = m + n$

Step 3: Place the ligands and lone pairs on a polyhedron of SN vertexes

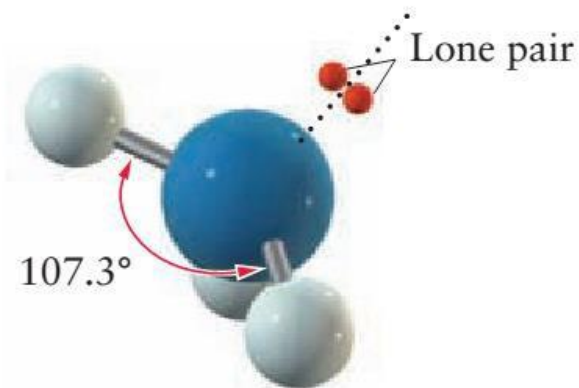
Step 4: Repulsion: $E-E > E-X > X-X$;

triple bond > double bond > single bond

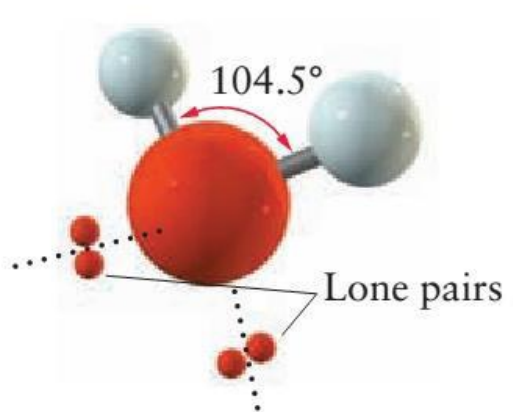
X high electronegativity (EN), less repulsion



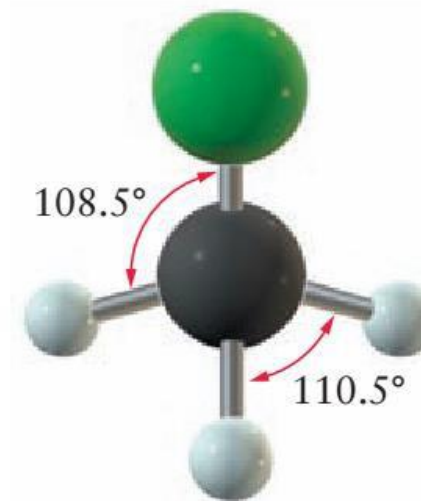
Molecular structures



NH_3



H_2O



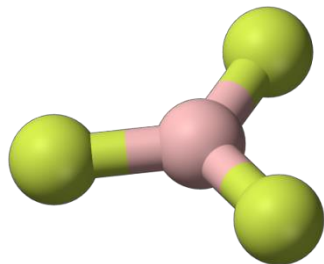
CH_3Cl

Step 4: Repulsion: $\text{E-E} > \text{E-X} > \text{X-X}$;

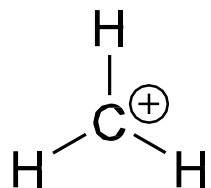
triple bond > double bond > single bond

X high electronegativity (EN), less repulsion

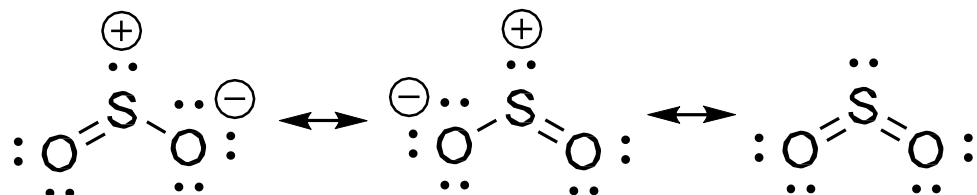
Example: $SN = 3$



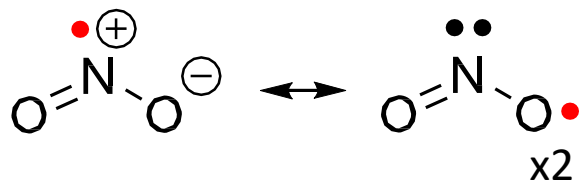
$BF_3: m = 3, n = 0$ Trigonal planar 平面三角形



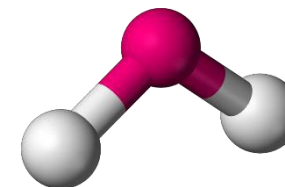
$CH_3^+: m = 3, n = 0$ Trigonal planar



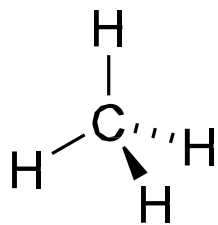
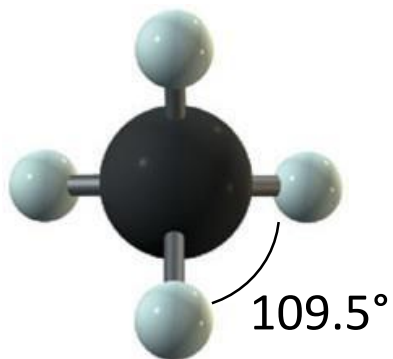
$SO_2: m = 2, n = 1$ Bent



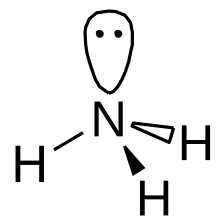
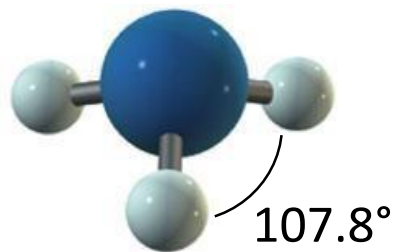
$NO_2: m = 2, 0.5 < n < 1$ Bent



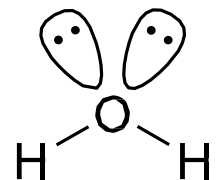
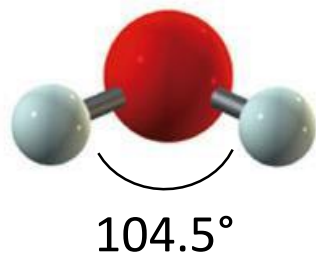
Example: SN = 4



CH_4 : $m = 4$, $n = 0$ Tetrahedral 四面体

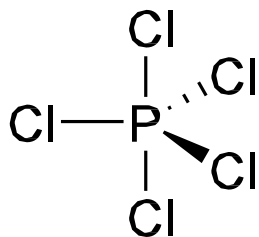


NH_3 : $m = 3$, $n = 1$ Trigonal pyramidal
三角锥

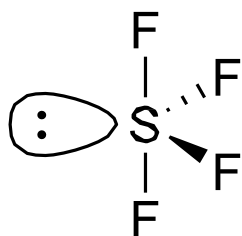
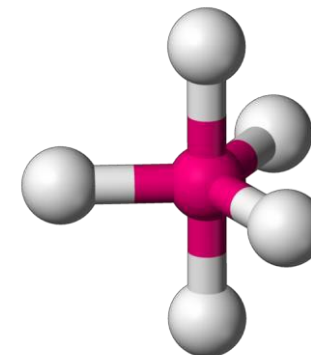


H_2O : $m = 2$, $n = 2$ Bent

Example: SN = 5

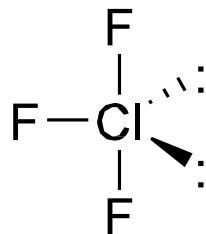
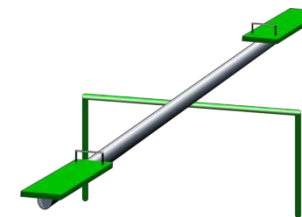


$\text{PCl}_5: m = 5, n = 0$ Trigonal bipyramidal
三角双锥

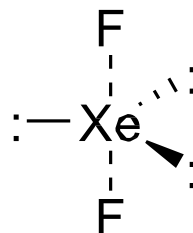
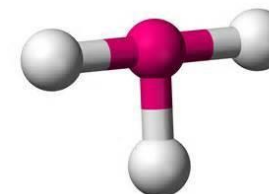


$\text{SF}_4: m = 4, n = 1$ Seesaw

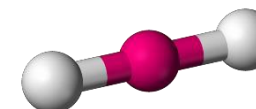
跷跷板



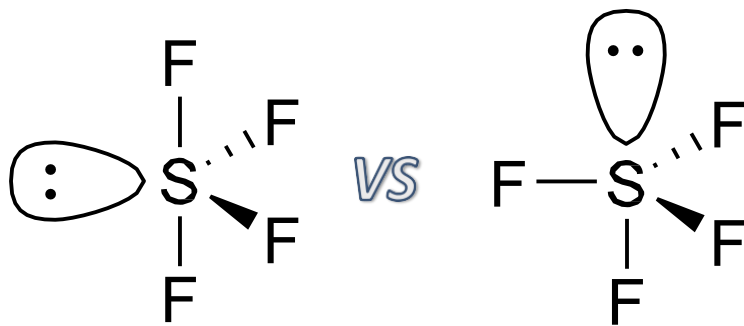
$\text{ClF}_3: m = 3, n = 2$ T-shaped



$\text{XeF}_2: m = 2, n = 3$ Linear

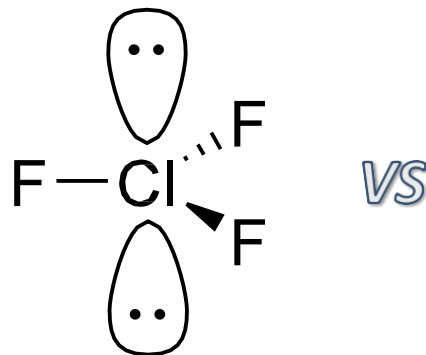
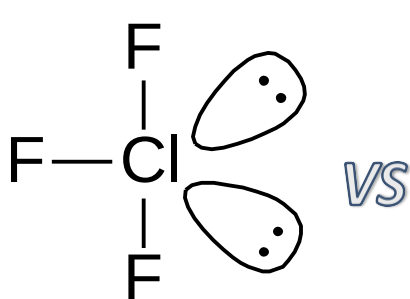


Lone electron pairs



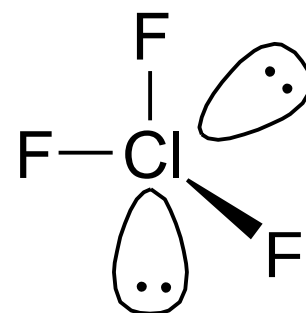
E-X 90° 2 pairs
X-X 90° 4 pairs

E-X 90° 3 pairs
X-X 90° 3 pairs



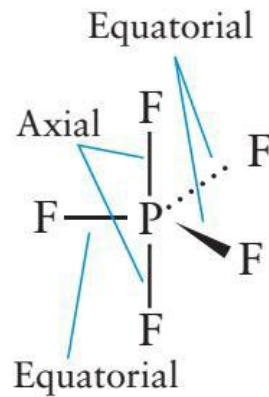
E-E 90° 0 pairs
E-X 90° 4 pairs
X-X 90° 2 pairs

E-E 90° 0 pairs
E-X 90° 6 pairs
X-X 90° 0 pairs



E-E 90° 1 pair
E-X 90° 3 pairs
X-X 90° 2 pairs

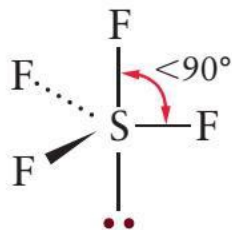
Molecular structures



(a) PF_5



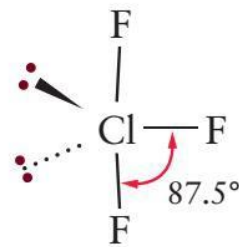
Trigonal bipyramid



(c) SF_4



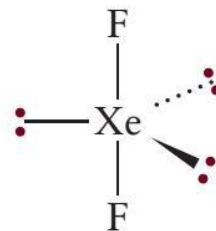
Distorted pyramid
(high-energy, not favored)



(d) ClF_3



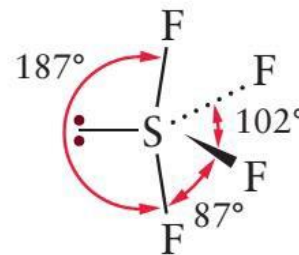
Distorted T



(e) XeF_2



Linear



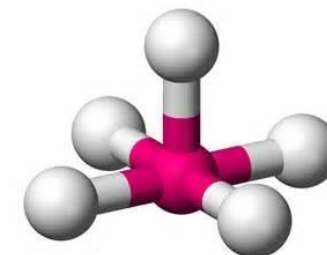
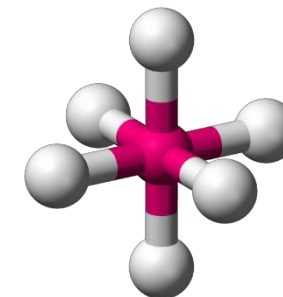
(b) SF_4



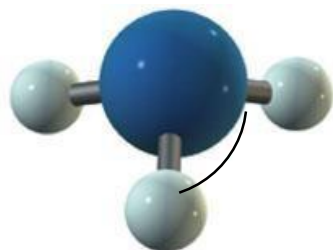
Seesaw
(low-energy, favored)

Summary of molecular geometry

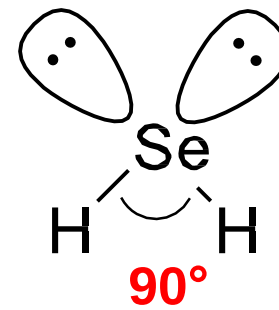
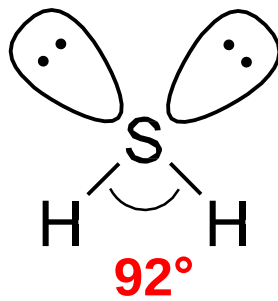
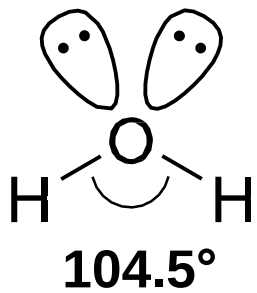
SN of A	Molecular type X is a bonded atom; E is a lone pair	Predicted shape
2	AX_2	Linear (线形)
3	AX_3	Trigonal planar (三角形)
	AX_2E	Bent (弯曲形)
4	AX_4	Tetrahedral (四面体)
	AX_3E	Trigonal pyramidal (三角锥)
	AX_2E_2	Bent
5	AX_5	Trigonal bipyramid (三角双锥)
	AX_4E	Distorted see-saw (跷跷板)
	AX_3E_2	Distorted T (T字)
	AX_2E_3	Linear
6	AX_6	Octahedral (八面体)
	AX_5E	Square pyramidal (四角锥)
	AX_4E_2	Square planar (正方平面)



Lone Pairs Are Fat

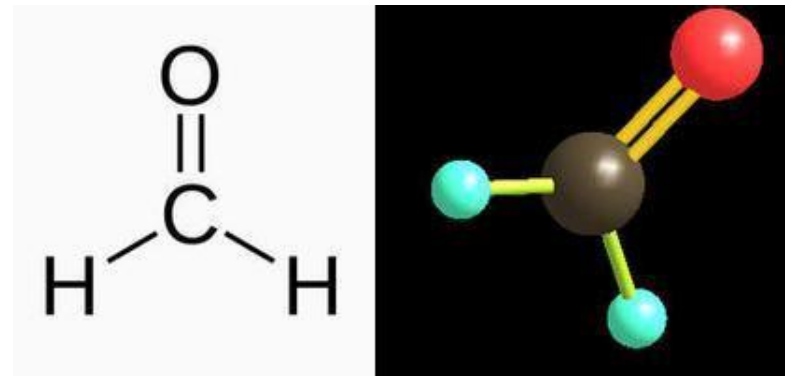


PCl_3	PBr_3	PI_3
100.1°	100.5°	102°



Double bond or triple bond are fat

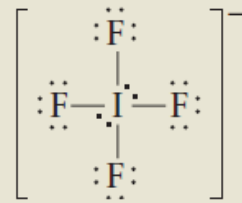
Molecules	Angle of XCX	Angle of XCO
$\text{F}_2\text{C}=\text{O}$	112.5°	123.2°
$\text{Cl}_2\text{C}=\text{O}$	111.3° (size effect)	124.3°
$\text{H}_2\text{C}=\text{O}$	118°	121°
$(\text{NH}_2)_2\text{C}=\text{O}$	118°	121°



Calculate steric numbers for iodine in IF_4^- and for bromine in BrO_4^- . These molecular ions have central I or Br surrounded by the other four atoms.

Solution

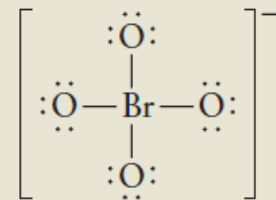
The central I^- atom has eight valence electrons.



Each F atom has seven valence electrons of its own and needs to share one of the electrons from I^- to achieve a noble-gas configuration. Thus, four of the I^- valence electrons take part in covalent bonds, leaving the remaining four to form two lone pairs. The steric number is given by

$$\text{SN} = 4 \text{ (bonded atoms)} + 2 \text{ (lone pairs)} = 6$$

In BrO_4^- , each oxygen atom needs to share two of the electrons from Br^- to achieve a noble-gas configuration.



Because this assignment accounts for all eight of the Br^- valence electrons, there are no lone pairs on the central atom and

$$\text{SN} = 4 \text{ (bonded atoms)} + 0 \text{ (lone pairs)} = 4$$

Predict the geometry of the following molecules and ions: (a) ClO_3^+ , (b) ClO_2^+ , (c) SiH_4 , (d) IF_5 .

Solution

- (a) The central Cl atom has all of its valence electrons (six, because the ion has a net positive charge) involved in bonds to the surrounding three oxygen atoms and has no lone pairs. Its steric number is 3. In the molecular ion, the central Cl should be surrounded by the three O atoms arranged in a trigonal planar structure.
- (b) The central Cl atom in this ion also has a steric number of 3, comprising two bonded atoms and a single lone pair. The predicted molecular geometry is a bent molecule with an angle somewhat less than 120° .
- (c) The central Si atom has a steric number of 4 and no lone pairs. The molecular geometry should consist of the Si atom surrounded by a regular tetrahedron of H atoms.
- (d) Iodine has seven valence electrons, of which five are shared in bonding pairs with F atoms. This leaves two electrons to form a lone pair, so the steric number is 5 (bonded atoms) + 1 (lone pair) = 6. The structure will be based on the octahedron of electron pairs from Figure 3.21, with five F atoms and one lone pair. The lone pair can be placed on any one of the six equivalent sites and will cause the four F atoms to bend away from it toward the fifth F atom, giving the distorted structure shown in Figure 3.24.

Predict whether the molecules NH_3 and SF_6 will have dipole moments.

Solution

The NH_3 molecule has a dipole moment because the three N—H bond dipoles add to give a net dipole pointing upward to the N atom from the base of the pyramid that defines the NH_3 structure. The SF_6 molecule has no dipole moment because each S—F bond dipole is balanced by one of equal magnitude pointing in the opposite direction on the other side of the molecule.

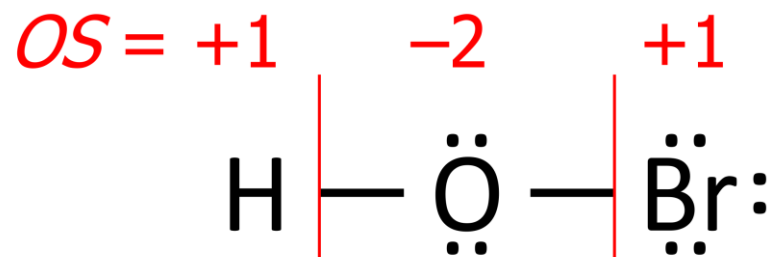
Major points of today's lecture

- **Atomic shells and classical models of chemical bonding**
 1. Forces and potential energy in molecules: formation of chemical bonds
 2. Ionic bonding
 3. Covalent and polar covalent bonding
 4. Electron pair bonds and Lewis diagrams for molecules
 5. The shapes of molecules
 6. Oxidation numbers
 7. Inorganic nomenclature

Oxidation numbers

Oxidation number (氧化数) describes the degree of oxidation (loss of electrons) of an atom in a chemical compound. Conceptually, the oxidation state, which may be positive, negative or zero

Oxidation number is the **hypothetical charge** that an atom would have if all bonds to atoms of different elements were 100% ionic, with no covalent component. This is never exactly true for real bonds.



Oxidation numbers

1. The oxidation numbers of the atoms in a neutral molecule must add up to zero, and those in an ion must add up to the charge on the ion. (charge conservation)
2. **Alkali-metal** atoms are assigned the **oxidation number +1**, and the **alkaline earth atoms +2**, in their compounds.
3. Fluorine is always assigned oxidation number -1. The other halogens are generally assigned oxidation number -1 in their compounds, **except** those containing oxygen and other halogens in which the halogen can have a positive oxidation number.

Oxidation numbers

4. **Hydrogen** is assigned oxidation number **+1** in its compounds, **except** in metal hydrides such as LiH, where the oxidation number of hydrogen is -1.

5. **Oxygen** is assigned oxidation number **-2** in nearly all compounds

Exceptions:

- In compounds with fluorine, the oxidation number for oxygen in OF_2 is +2 as F is always -1.
- In peroxides (such as H_2O_2 and Na_2O_2), the oxidation number for oxygen is -1 as H and Na is always +1
- In superoxides (such as KO_2), the oxidation number for oxygen is $-\frac{1}{2}$.

Oxidation numbers

T A B L E 3.9 Formulas and Names of Some Common Anions

F^-	Fluoride	CO_3^{2-}	Carbonate
Cl^-	Chloride	HCO_3^-	Hydrogen carbonate
Br^-	Bromide	NO_2^-	Nitrite
I^-	Iodide	NO_3^-	Nitrate
H^-	Hydride	SiO_4^{4-}	Silicate
O^{2-}	Oxide	PO_4^{3-}	Phosphate
S^{2-}	Sulfide	HPO_4^{2-}	Hydrogen phosphate
O_2^{2-}	Peroxide	$H_2PO_4^-$	Dihydrogen phosphate
O_2^-	Superoxide	SO_3^{2-}	Sulfite
OH^-	Hydroxide	SO_4^{2-}	Sulfate
CN^-	Cyanide	HSO_4^-	Hydrogen sulfate
CNO^-	Cyanate	ClO^-	Hypochlorite
SCN^-	Thiocyanate	ClO_2^-	Chlorite
MnO_4^-	Permanganate	ClO_3^-	Chlorate
CrO_4^{2-}	Chromate	ClO_4^-	Perchlorate
$Cr_2O_7^{2-}$	Dichromate		

In Ag_2S , sulfur appears as the sulfide ion and has oxidation number -2, but in Ag_2SO_4 , it appears as part of a sulfate (SO_4^{2-}) ion. In this case,

$$(\text{oxidation number of S}) + [4 \times (\text{oxidation number of O})] = \text{total charge on ion}$$

$$x + [4(-2)] = -2$$

$$\text{oxidation number of S} = x = +6$$

Oxidation numbers

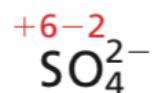
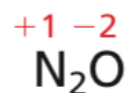
Most elements have more than one possible oxidation state. For example, carbon has nine possible integer oxidation states from -4 to $+4$:

Integer oxidation states of carbon

Oxidation state	Example compound
-4	CH_4
-3	C_2H_6
-2	C_2H_4 , CH_3Cl
-1	C_2H_2 , C_6H_6 , $(\text{CH}_2\text{OH})_2$
0	HCHO , CH_2Cl_2
$+1$	OCHCHO , $\text{CHCl}_2\text{CHCl}_2$
$+2$	HCOOH , CHCl_3
$+3$	HOCCOOH , C_2Cl_6
$+4$	CCl_4 , CO_2

Oxidation numbers

A convenient way to indicate the oxidation number of an atom is to write it directly above the corresponding symbol in the formula of the compound:



Assign oxidation numbers to the atoms in the following chemical compounds and ions: NaCl, ClO^- , $\text{Fe}_2(\text{SO}_4)_3$, SO_2 , I_2 , KMnO_4 , CaH_2 .

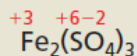
Solution



From conventions 2 and 3.



From conventions 1 and 5.



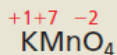
From conventions 1 and 5. This is solved by recognizing the presence of sulfate SO_4^{2-} groups.



From conventions 1 and 5.



From convention 1 (I_2 is an element).



From conventions 1, 2, and 5.



From conventions 1 and 2 (metal hydride case).

Oxidation numbers

- **Formal charges** are used solely to identify preferred Lewis diagrams.
- **Oxidation numbers** are used in **nomenclature**, in **identifying oxidation-reduction reactions**, and in exploring trends in chemical reactivity across the periodic table.
- Oxidation numbers are often not the same as the true charges on atoms.



Mn: + 7

A high oxidation state usually indicates significant covalent character in the bonding of that compound.

Major points of today's lecture

- **Atomic shells and classical models of chemical bonding**
 1. Forces and potential energy in molecules: formation of chemical bonds
 2. Ionic bonding
 3. Covalent and polar covalent bonding
 4. Electron pair bonds and Lewis diagrams for molecules
 5. The shapes of molecules
 6. Oxidation numbers
 7. Inorganic nomenclature

Names and formulas of ionic compounds

An ionic compound is named by listing the name of the cation followed by that of the anion.

1-1. A monatomic cation bears the name of the parent element.

Na^+ sodium ion; Ca^{2+} calcium ion

If a metal forms ions of more than one charge, placing a Roman numeral in parentheses after the name of the metal

Cu^+	copper(I) ion	Fe^{2+}	iron(II) ion	Sn^{2+}	tin(II) ion
Cu^{2+}	copper(II) ion	Fe^{3+}	iron(III) ion	Sn^{4+}	tin(IV) ion

Earlier method: suffixes -ous and -ic added to the root of the (usually Latin) name of the metal to indicate the ions of lower and higher charge

Fe^{2+} ferrous ion; Fe^3 ferric ion

1-2. Polyatomic cations

ammonium ion, NH_4^+ (obtained by adding H to ammonia)
hydronium ion, H_3O^+

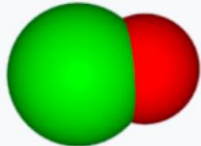
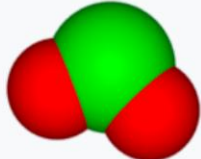
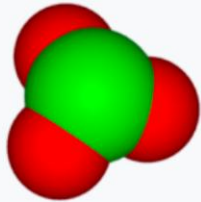
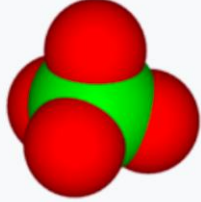
Names and formulas of ionic compounds

2-1. A monatomic anion is named by adding the suffix -ide to the first portion of the name of the element

chlorine becomes the chloride ion; oxygen becomes the oxide ion

2-2. Polyatomic anions

An oxoanion, is an ion with the generic formula $A_xO_y^{z-}$

Oxidation state	Name	Formula	Image
+1	The hypochlorite ion	ClO^-	
+3	The chlorite ion	ClO_2^-	
+5	The chlorate ion	ClO_3^-	
+7	The perchlorate ion	ClO_4^-	

Names and formulas of ionic compounds

2-2. Polyatomic anions

If central atom is in Group VII

Central atom oxidation number	Naming scheme	Examples
= Group number	per-*-ate	Perchlorate (ClO_4^-), Perbromate (BrO_4^-), Periodate (IO_4^-), Permanganate (MnO_4^-)
= Group number – 2	*-ate	Chlorate (ClO_3^-), Bromate (BrO_3^-), Iodate (IO_3^-)
= Group number – 4	*-ite	Chlorite (ClO_2^-), Bromite (BrO_2^-)
= Group number – 6	hypo-*-ite	Hypochlorite (ClO^-), Hypobromite (BrO^-)

Names and formulas of ionic compounds

2-2. Polyatomic anions

Central atom oxidation number	Naming scheme	Examples
= Group number	*-ate	Borate (BO_3^{3-}), Carbonate (CO_3^{2-}), Nitrate (NO_3^-), Phosphate (PO_4^{3-}), Sulfate (SO_4^{2-}), Chromate (CrO_4^{2-}), Arsenate (AsO_4^{3-}), Ferrate (FeO_4^{2-})
= Group number – 2	*-ite	Nitrite (NO_2^-), Phosphite (PO_3^{3-}), Sulfite (SO_3^{2-}), Arsenite (AsO_3^{3-})
= Group number – 4	hypo-*-ite	Hypophosphite (PO_2^{3-}), Hyposulfite (SO_2^{2-})

Naming binary covalent compounds

If a pair of elements forms **only one compound**, begin with the name of the element that appears first in the chemical formula, followed by the second element, with the suffix **-ide** added to its root.

HBr	hydrogen bromide
BeCl ₂	beryllium chloride
H ₂ S	hydrogen sulfide
BN	boron nitride

Well-established nonsystematic names continues to be used.

H ₂ O	water
NH ₃	ammonia
N ₂ H ₄	hydrazine
PH ₃	phosphine
AsH ₃	arsine
COCl ₂	phosgene

Naming binary covalent compounds

If a pair of elements forms more than one compound, two methods can be used:

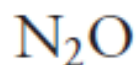
1. Use Greek prefixes to specify the number of atoms of each element. If the compound is a solid without well-defined molecules, name the empirical formula.

2. Write the oxidation number of the first-named element in Roman numerals and place it in parentheses after the name of that element.

TABLE 3.10

Prefixes Used for Naming Binary Covalent Compounds

Number	Prefix
1	<i>mono-</i>
2	<i>di-</i>
3	<i>tri-</i>
4	<i>tetra-</i>
5	<i>penta-</i>
6	<i>hexa-</i>
7	<i>hepta-</i>
8	<i>octa-</i>
9	<i>nona-</i>
10	<i>deca-</i>



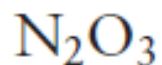
dinitrogen oxide

nitrogen(I) oxide



nitrogen oxide

nitrogen(II) oxide



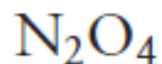
dinitrogen trioxide

nitrogen(III) oxide



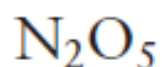
nitrogen dioxide

nitrogen(IV) oxide



dinitrogen tetroxide

nitrogen(IV) oxide



dinitrogen pentoxide

nitrogen(V) oxide

The prefix for one (mono-) is omitted, except in the case of carbon monoxide.

Give the chemical formulas of (a) calcium cyanide and (b) copper(II) phosphate.

Solution

- (a) Calcium cyanide is composed of Ca^{2+} and CN^- . For the overall charge to be 0, there must be two CN^- for each Ca^{2+} . Thus, the chemical formula of calcium cyanide is $\text{Ca}(\text{CN})_2$.
- (b) The ions present in copper(II) phosphate are Cu^{2+} and PO_4^{3-} . To ensure charge neutrality, there must be three Cu^{2+} (total charge +6) and two PO_4^{3-} (total charge -6) per formula unit. Thus, the chemical formula of copper(II) phosphate is $\text{Cu}_3(\text{PO}_4)_2$.

Homework: read “principle of modern chemistry”, chapter 3.